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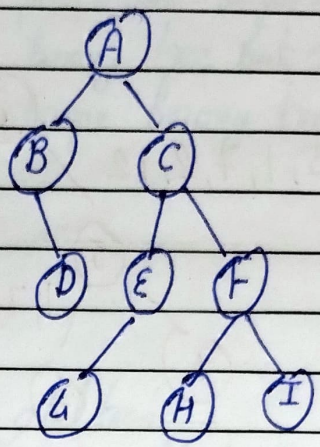
Inorder, Preorder and Postorder traversal of a binary tree

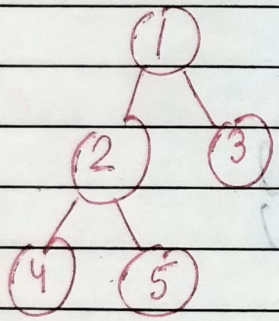
Inorder:- Left Root Right
Preorder- Root Left Right
Postorder- Left Right Root

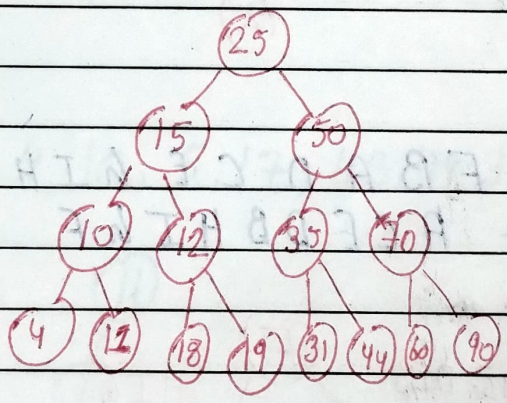


Inorder:-
BDAGECHFI

Preorder:- ABDCEGFHI

Postorder:- DBGEHIFCA





Preorder:- 1 2 4 5 3
Postorder:- 4 5 2 3 1
Inorder: 4 2 5 1 3

Postorder:- 4 11 10 18 19 12 15 31 44 35 60 90 70 50 25
Preorder:- 25 15 10 4 11 12 18 19 50 35 31 44 70 60 90
Inorder:- 4 10 11 15 18 12 19 25 31 35 44 50 60 70 90

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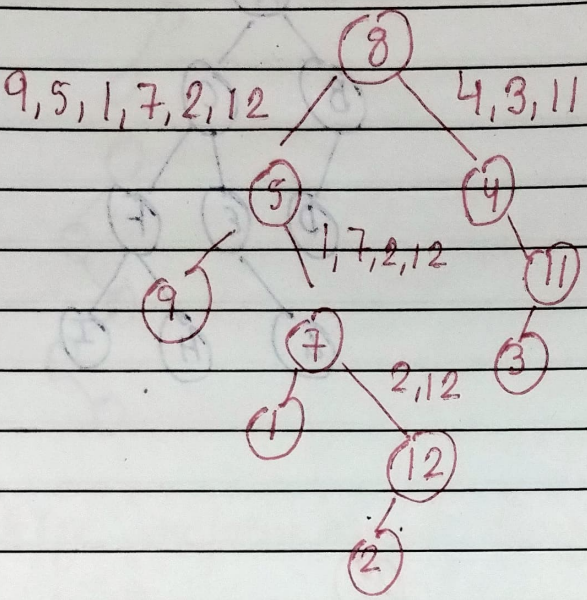
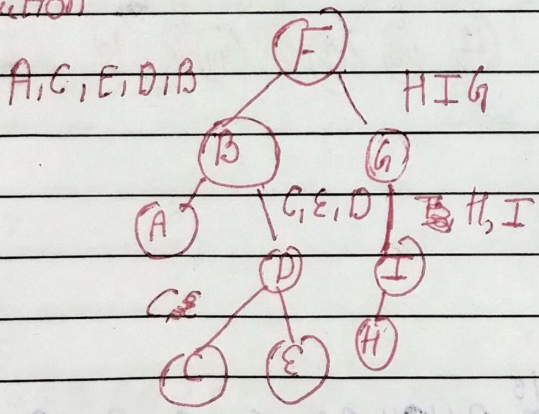
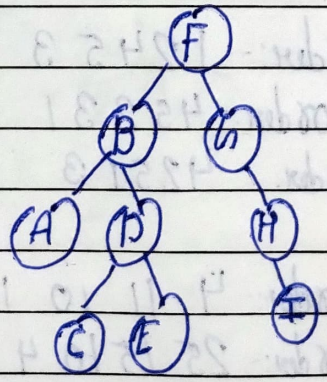
Postorder:- 9, 1, 2, 12, 7, 5, 3, 11, 4, 8 — L R Root
Inorder - 9, 5, 1, 7, 2, 12, 8, 4, 3, 11 — L Root R



Preorder:- F B A D C E G I H (Root L R)
Postorder - A C E D B H I G F (L R Root)

I Solution





Preorder:- F B A D C E G I H
Postorder:- A C E D B H I G F

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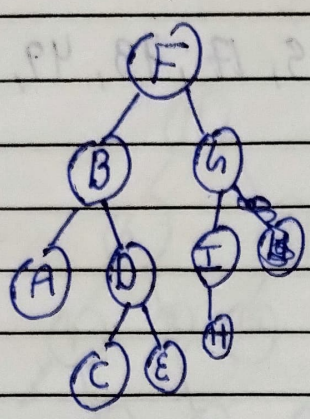
II - Solution



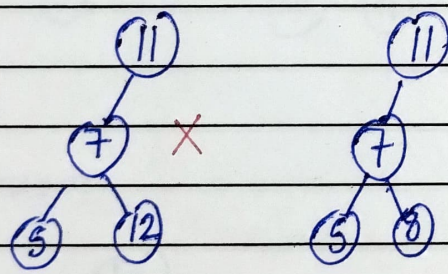
Using preorder and postorder unique full binary tree but not unique binary tree.

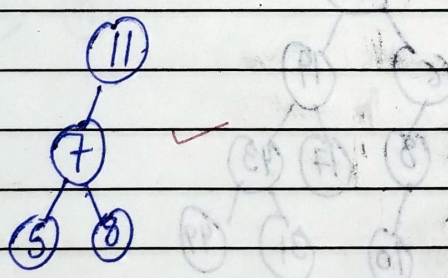
Binary Search Tree

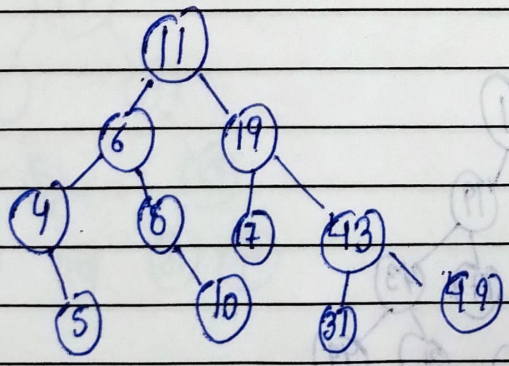
(1) Each node must have atmost 2 children
(2) The value of left node must be smaller than parent node and value of right node must be greater than parents.

Draw Binary Search tree

11, 6, 8, 19, 4, 10, 5, 17, 43, 49, 31







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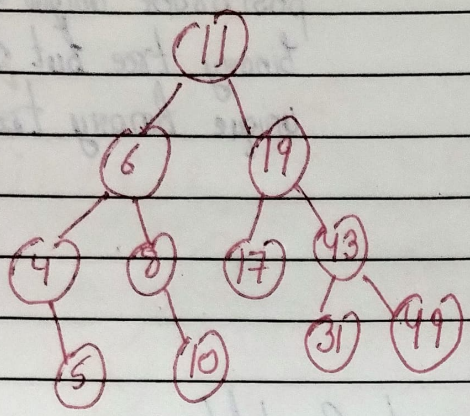
Deletion in BST

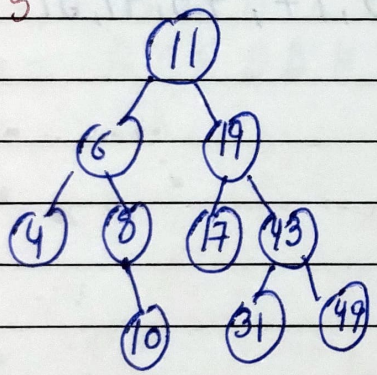
11, 6, 8, 19, 4, 10, 5, 17, 43, 49, 31

(1)

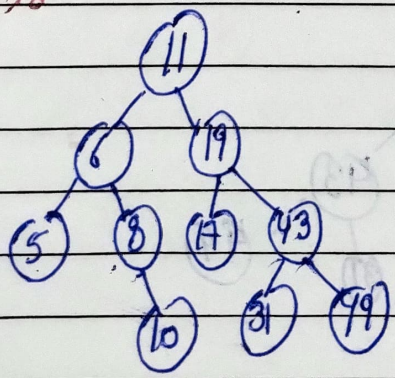


Case I No child
Case II 1 child
Case III 2 child

Case I No child
Delete 5



Case II 1 child
Delete 4



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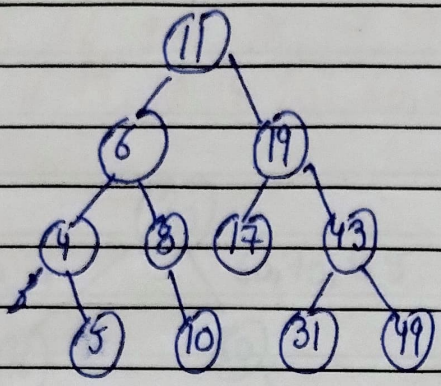
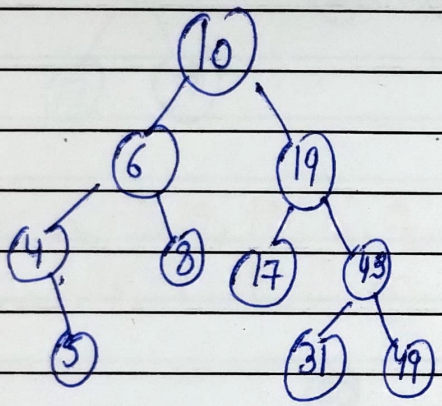
Case III 2 child (i) Inorder Predecessor (ii) Inorder Successor

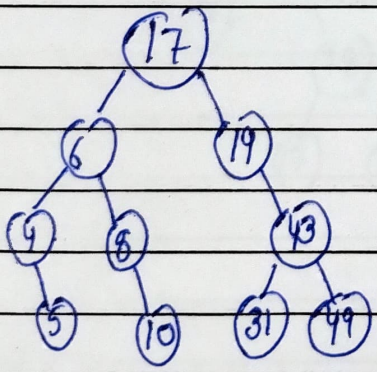
Delete (11)



Inorder Predecessor



Inorder Successor



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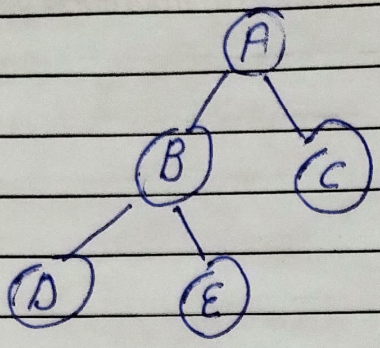
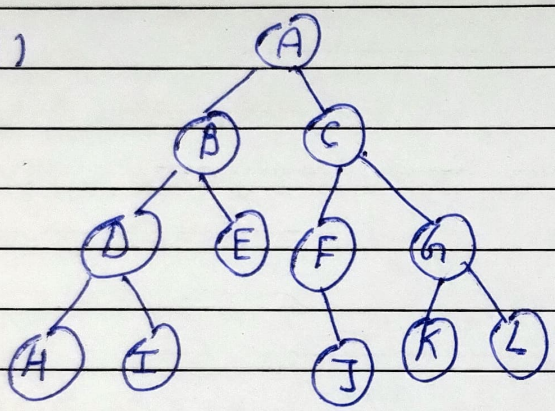
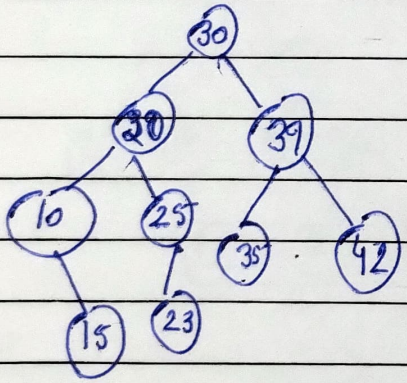
(1)



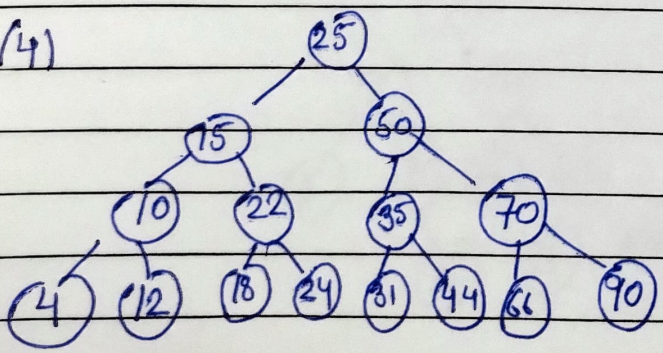
(2)



(3)



(4)



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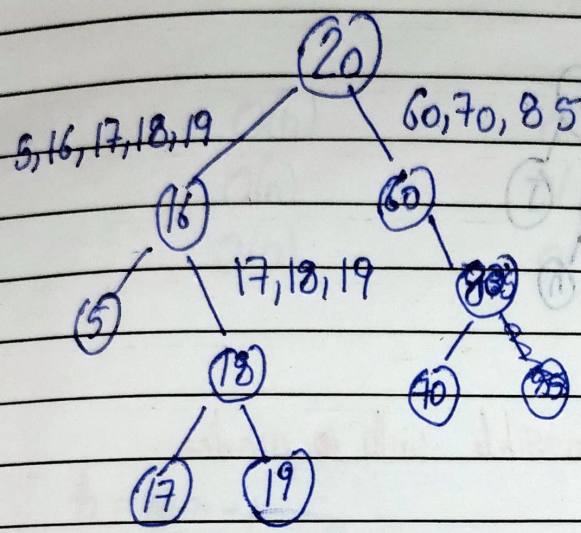
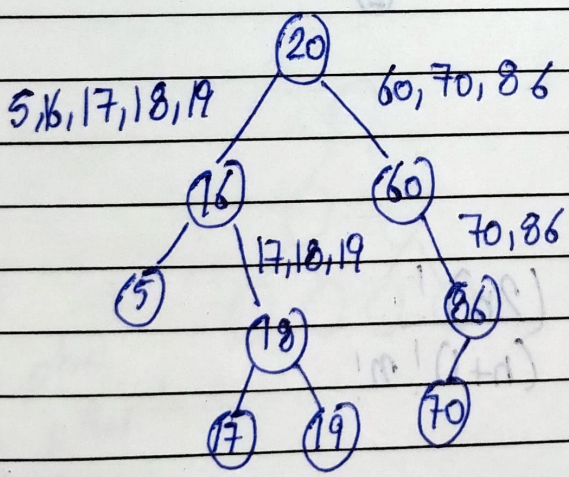
Construction of BST when only preorder or postorder is given

Preorder :- 20, 16, 5, 18, 17, 19, 60, 85, 70 (Root left Right)
Inorder :- 5, 16, 17, 18, 19, 20, 60, 70, 85 (left Root Right)



Postorder :- 5, 17, 19, 18, 16, 70, 86, 60, 20 (left Right Root)
Inorder :- 5, 16, 17, 18, 19, 20, 60, 70, 86



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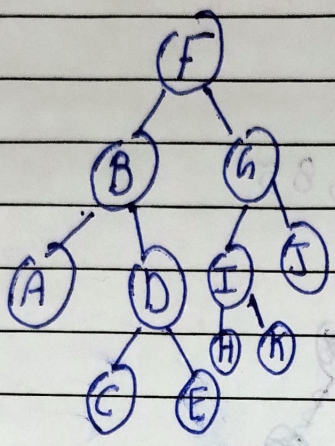
Construct Binary tree from given preorder & postorder

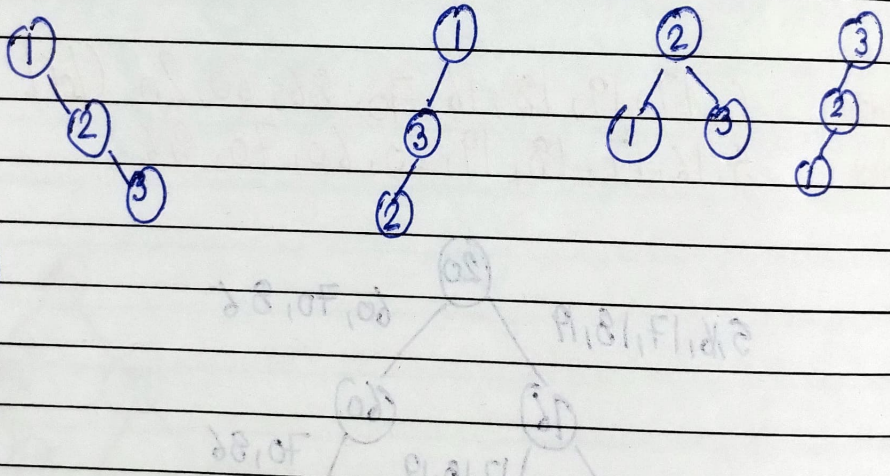
Preorder - F B A D C E G I H K J (Root L R)

Postorder - A C E D B H K J I G F (L R Root)



Binary ~~tree~~ Search tree possible with 3 nodes.

1, 2, 3



$$C_n = \frac{(2n)!}{(n+1)!\,n!}$$

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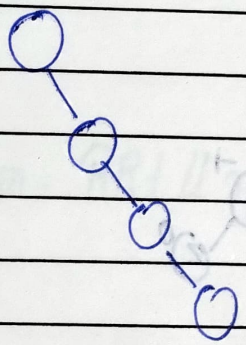
Time and space complexity.
Binary Search tree

Operation	Worst case	Average case	Best case	Space
Search	O(n)	O(logn)	~~O(logn)~~ O(1)	O(n)
Insert	O(n)	O(logn)	~~O(logn)~~ O(1)	O(n)
Delete	O(n)	O(logn)	O(logn)	O(n)

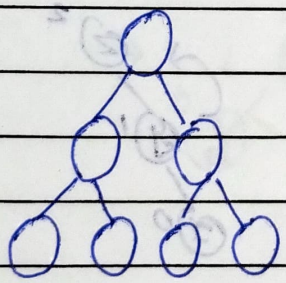
Worst case



$h = n-1$

Time complexity = O(n)

Average case



$n = 2^{h+1} - 1$

$n+1 = 2^{h+1}$

$\log_2(n+1) = (h+1)\log_2 2$

$h+1 = \log_2(n+1)$

$\boxed{h = \log_2(n+1) - 1}$

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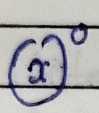
AVL Tree

① It is a Binary Search tree.

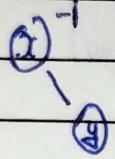
② Height of left Subtree - Height of right Subtree = {-1, 0, 1} called Balance factor.

Case I



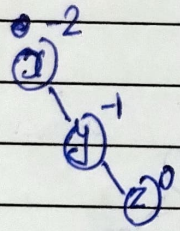
Balance tree

Case II



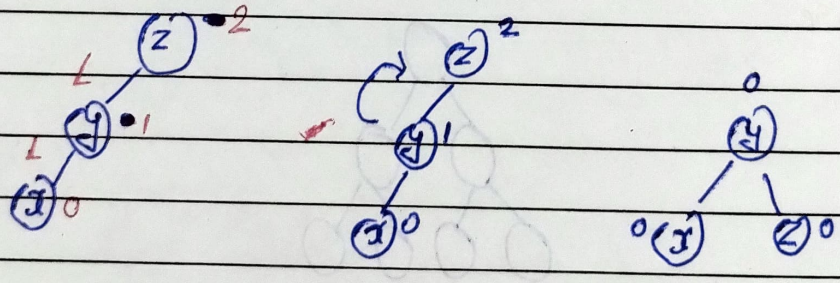
Balanced tree

Case III

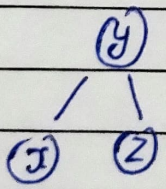


Unbalanced tree

(a) LL Rotation



Clockwise rotation



Balanced tree

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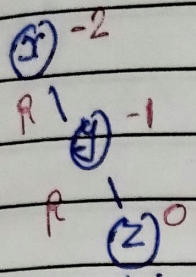
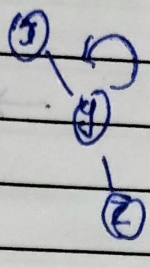
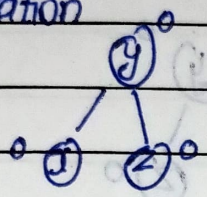
(b) RR rotation

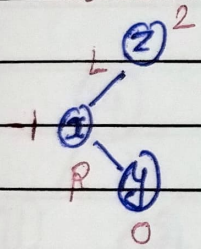




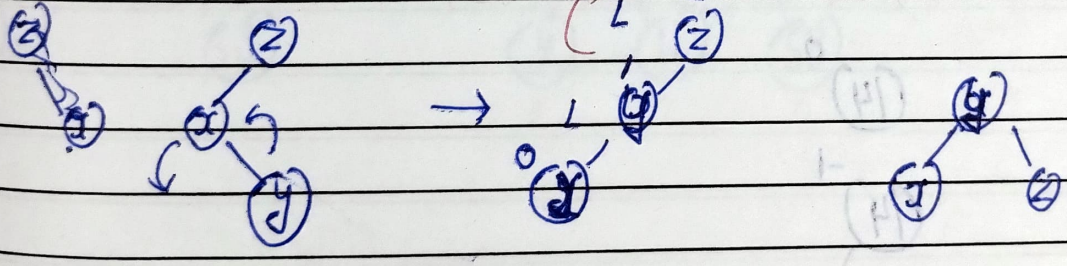
Anticlockwise rotation



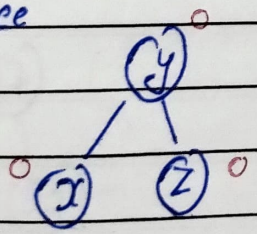
(c) LR rotation



LR rotation = RR + LL rotation



Balanced tree



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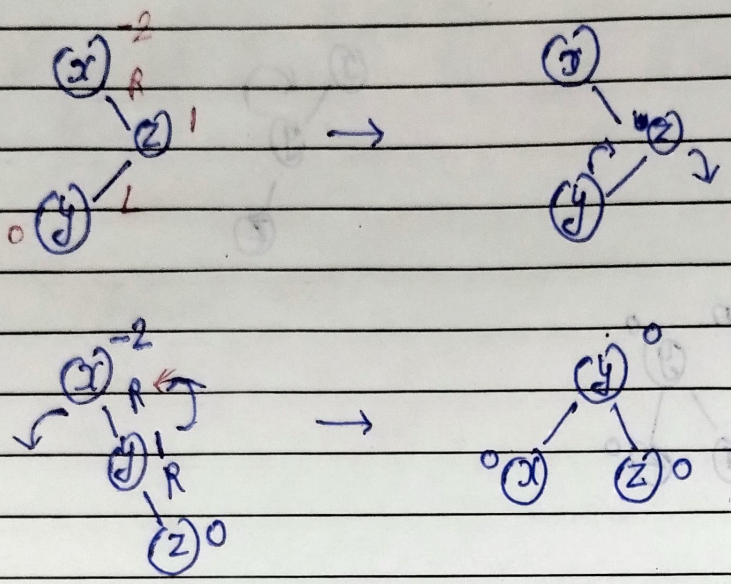
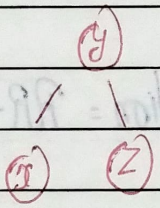
(d) RL rotation

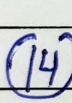


Balanced tree

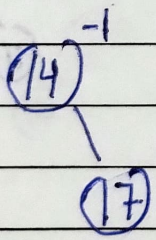


Eg- 14, 17, 11, 7, 53, 4, 13, 12, 8, 60, 19, 16, 20

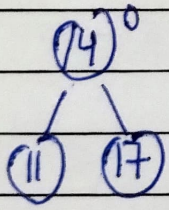
I



II



III



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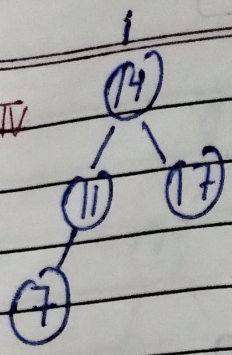
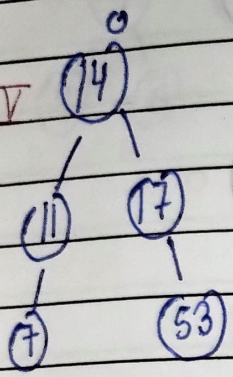
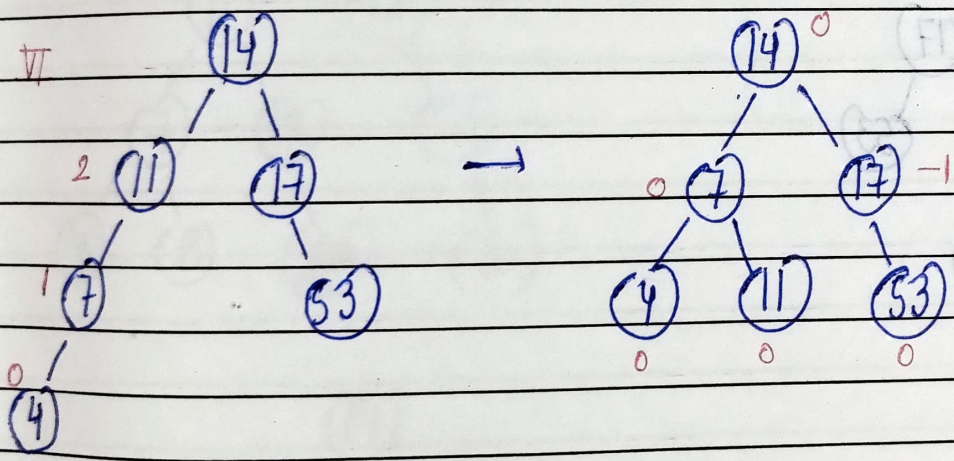
IV



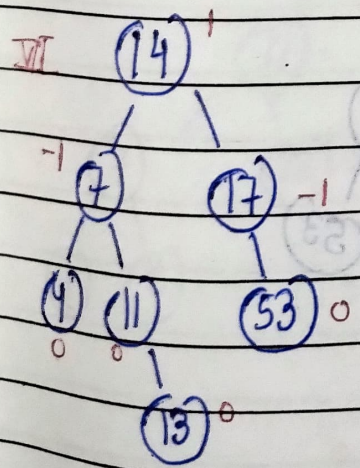
V



VI



VII



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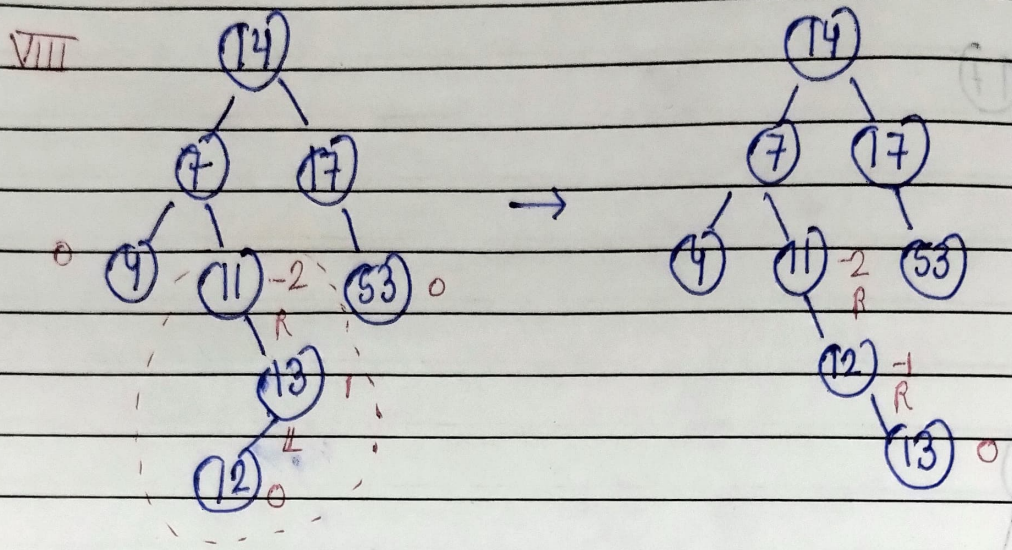
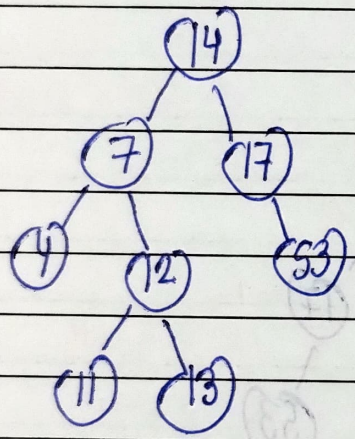
VIII





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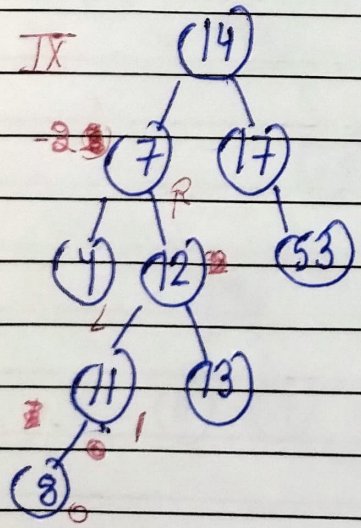

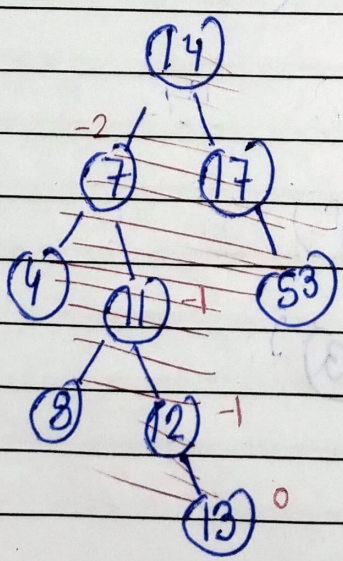

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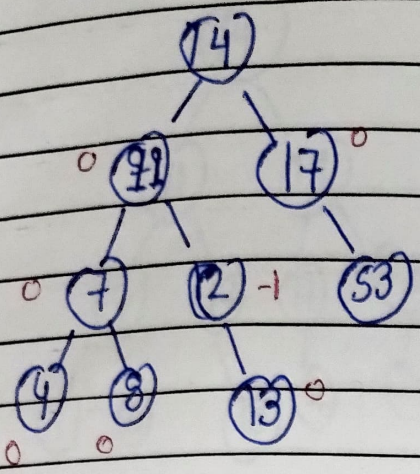

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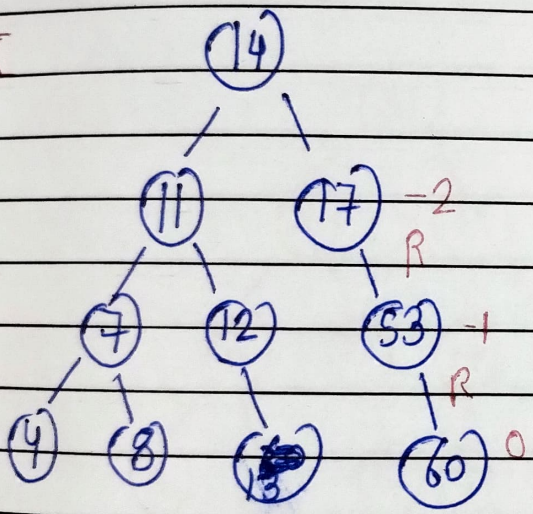

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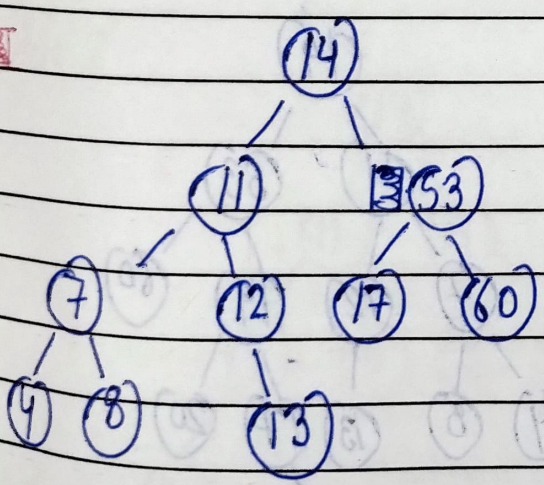

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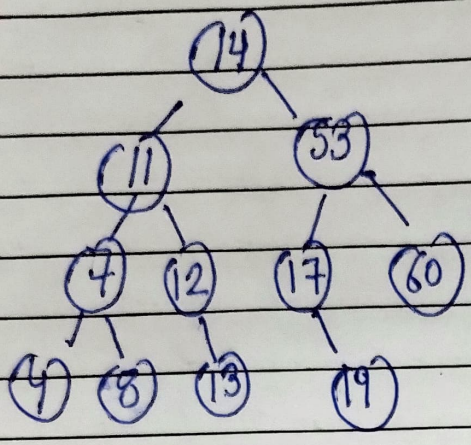
XI



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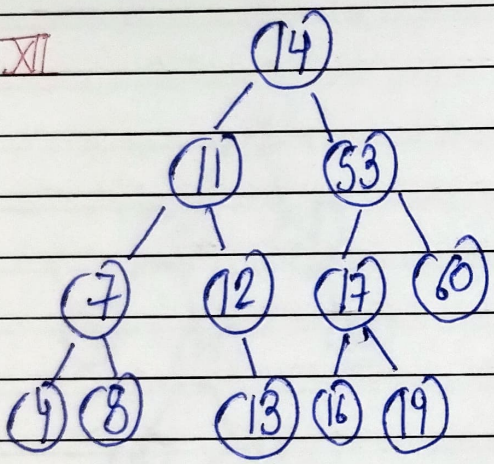

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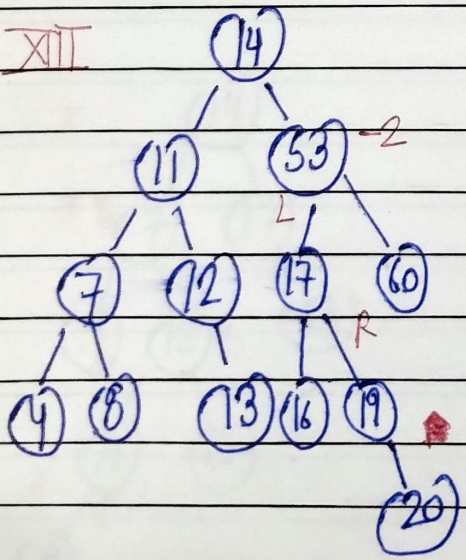

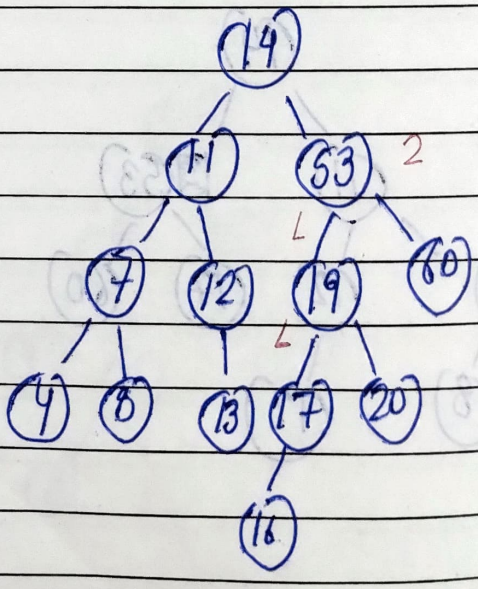

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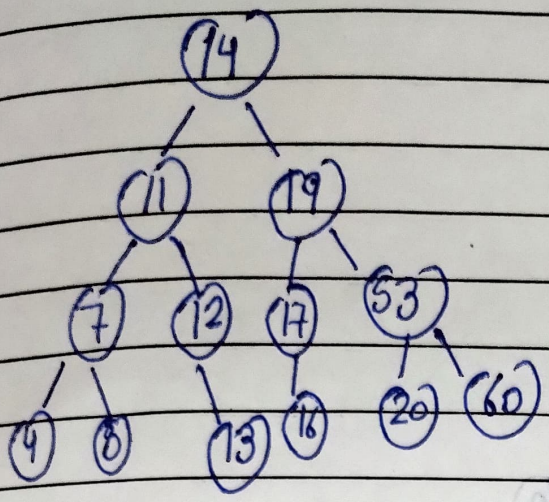
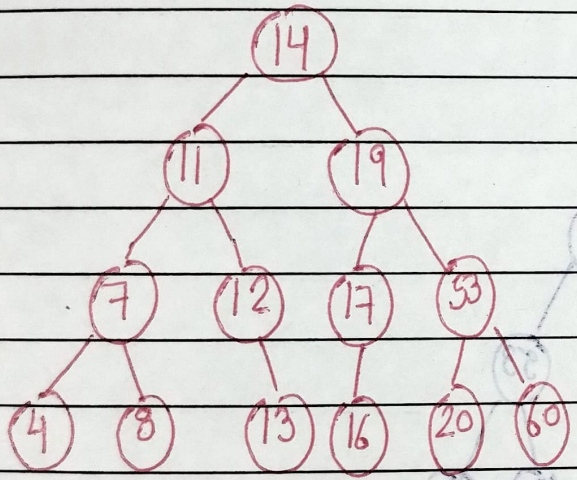
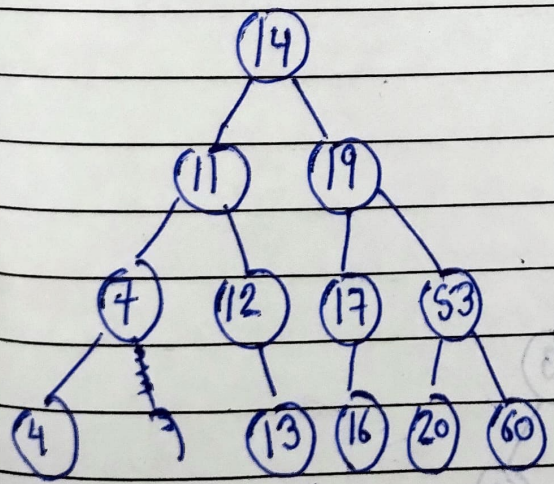


Deletion in AVL Tree

Step I Delete the node according to (BST cases)
Step II Calculate balance factor



Delete 8



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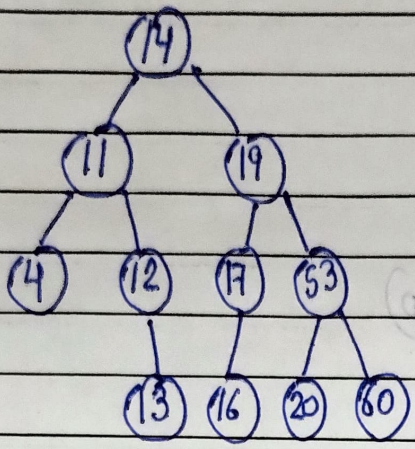

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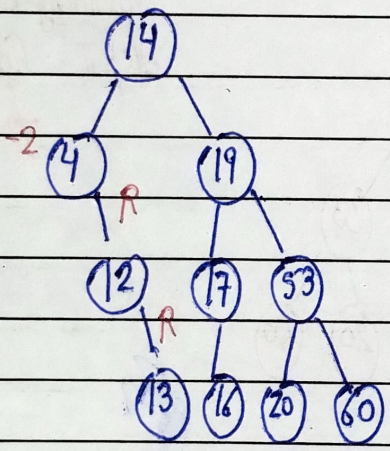

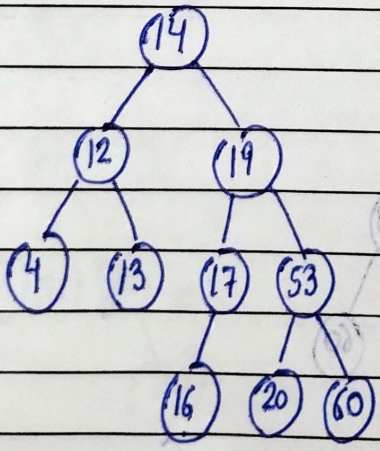

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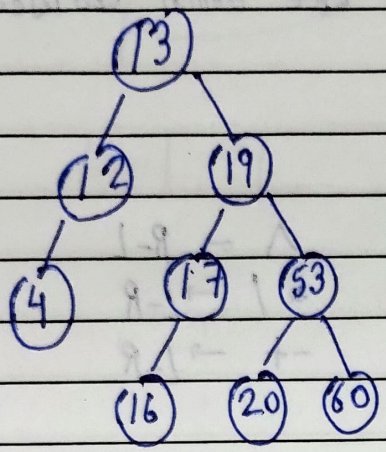

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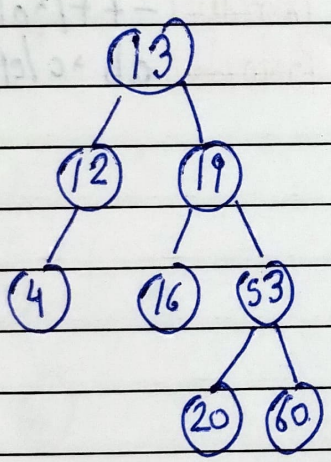

Time Complexity & Space Complexity

	Best case	Average case	Worst case
Insert	O(logn)	O(logn)	O(logn)
Delete	O(logn)	O(logn)	O(logn)
Search	O(1)	O(logn)	O(logn)
Traversal	O(logn)	O(logn)	O(logn)
Space	O(n)	O(n)	O(n)